Analysing E-Learning and Purchases using MySQL

Objectives:

To design and manage a relational **MySQL database** for an online learning platform and analyze learner purchase data to generate meaningful business insights. The project aims to understand **sales trends, learner purchasing behavior, course performance, and popular categories** using SQL queries, joins, aggregate functions, and data analysis techniques**.**

Dataset Description:

You will work with the following three tables:

Table: learners

Column	**Description**
learner_id	Primary Key
full_name	Learner name
country	Country of residence

Table: courses

Column	**Description**
course_id	Primary Key
course_name	Course title
category	Course category
unit_price	Course category

Table: purchases

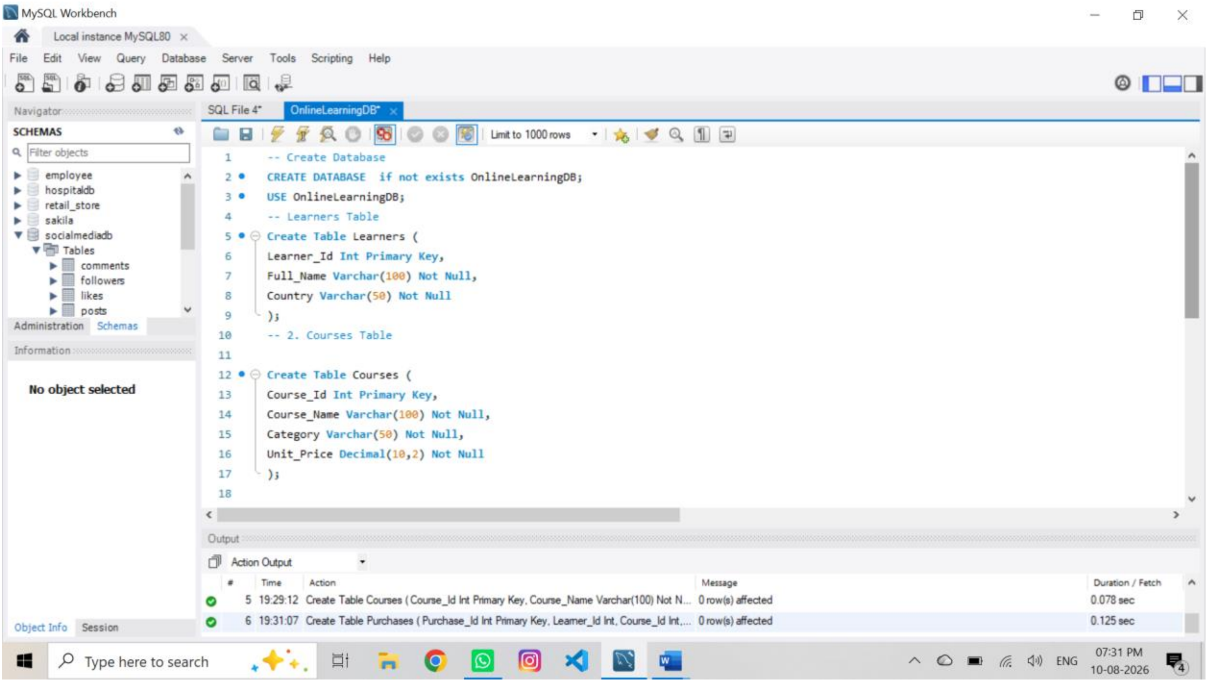
Columns	**Description**
purchase_id	Primary Key
learner_id	Foreign Key → learners
course_id	Foreign Key → learners
quantity	Number of courses purchased
purchase_date	Date of purchase

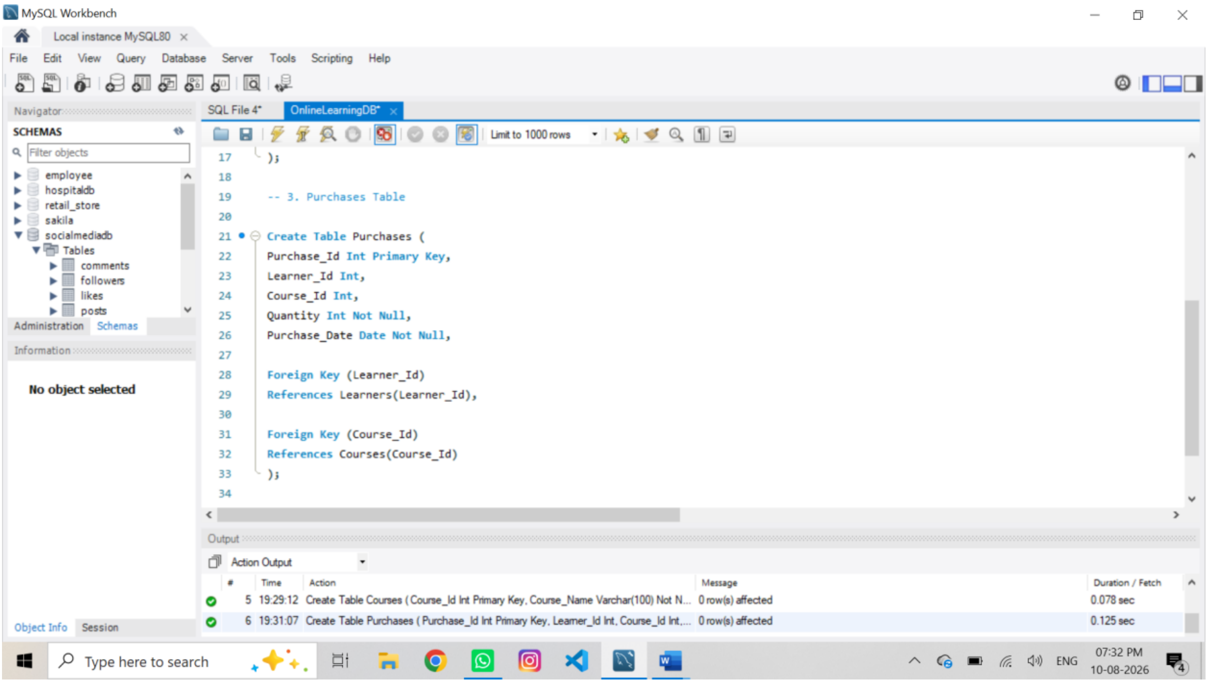
1.Database Setup & Data Entry

Create a new database.

- **Create the three tables with proper:**

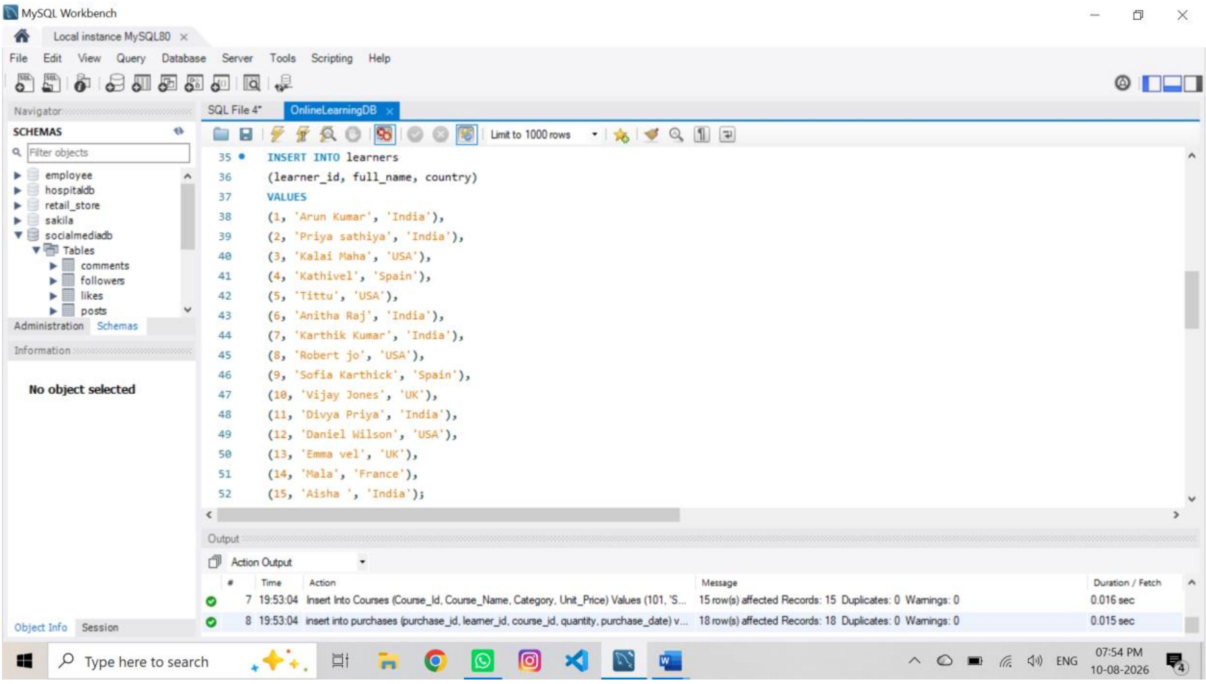
○ **Primary keys, Foreign keys, Appropriate data types**

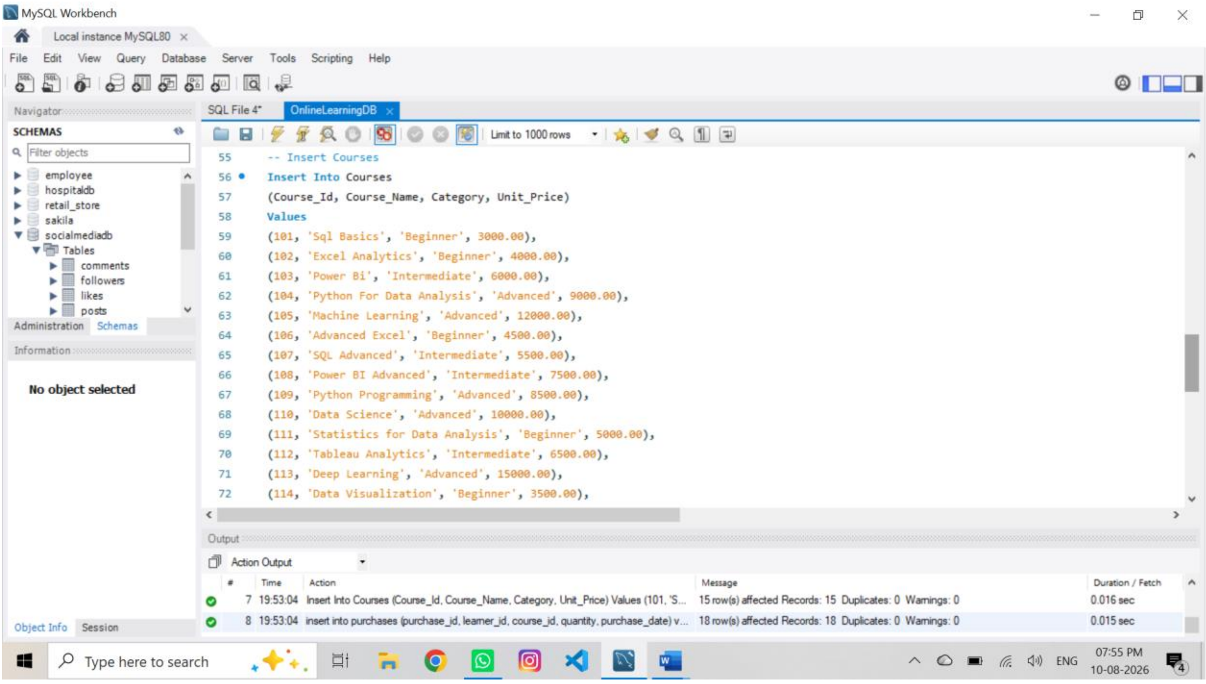




- **Insert sample data:**

○ **4–5 learners** ○ **4–5 courses (different categories)** ○ **6–8 purchase records**


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL File 4*
OnlineLearningDB
Limit to 1000 rows
Navigator
SCHEMAS
employee
hospitaldb
retail_store
sakila
socialmediadb
Tables
comments
followers
likes
posts
Administration Schemas
Information
No object selected
Object Info Session
INSERT INTO learners
(learner_id, full_name, country)
VALUES
(1, 'Arun Kumar', 'India'),
(2, 'Priya sathiya', 'India'),
(3, 'Kalai Maha', 'USA'),
(4, 'Kathivel', 'Spain'),
(5, 'Tittu', 'USA'),
(6, 'Anitha Raj', 'India'),
(7, 'Karthik Kumar', 'India'),
(8, 'Robert jo', 'USA'),
(9, 'Sofia Karthick', 'Spain'),
(10, 'Vijay Jones', 'UK'),
(11, 'Divya Priya', 'India'),
(12, 'Daniel Wilson', 'USA'),
(13, 'Emma vel', 'UK'),
(14, 'Mala', 'France'),
(15, 'Aisha ', 'India');
Output
Action Output
Time Action Message Duration / Fetch
7 19:53:04 Insert Into Courses (Course_Id, Course_Name, Category, Unit_Price) Values (101, 'S... 15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0 0.016 sec
8 19:53:04 insert into purchases (purchase_id, learner_id, course_id, quantity, purchase_date) v... 18 row(s) affected Records: 18 Duplicates: 0 Warnings: 0 0.015 sec
Type here to search
ENG
07:54 PM
10-08-2026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sakila
socialmediadb
Tables
comments
followers
likes
posts
Administration Schemas
Information
No object selected
Object Info Session
-- Insert Courses
Insert Into Courses
(Course_Id, Course_Name, Category, Unit_Price)
Values
(101, 'Sql Basics', 'Beginner', 3000.00),
(102, 'Excel Analytics', 'Beginner', 4000.00),
(103, 'Power Bi', 'Intermediate', 6000.00),
(104, 'Python For Data Analysis', 'Advanced', 9000.00),
(105, 'Machine Learning', 'Advanced', 12000.00),
(106, 'Advanced Excel', 'Beginner', 4500.00),
(107, 'SQL Advanced', 'Intermediate', 5500.00),
(108, 'Power BI Advanced', 'Intermediate', 7500.00),
(109, 'Python Programming', 'Advanced', 8500.00),
(110, 'Data Science', 'Advanced', 10000.00),
(111, 'Statistics for Data Analysis', 'Beginner', 5000.00),
(112, 'Tableau Analytics', 'Intermediate', 6500.00),
(113, 'Deep Learning', 'Advanced', 15000.00),
(114, 'Data Visualization', 'Beginner', 3500.00),
Output
Action Output
Time Action Message Duration / Fetch
7 19:53:04 Insert Into Courses (Course_Id, Course_Name, Category, Unit_Price) Values (101, 'S... 15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0 0.016 sec
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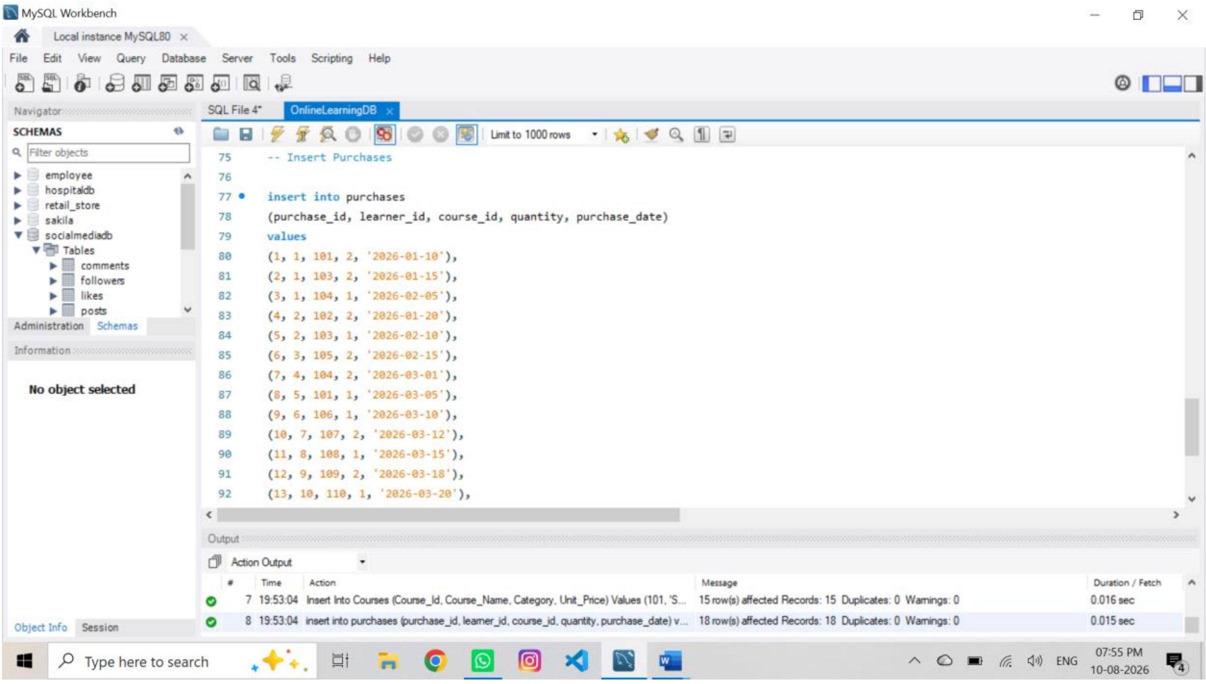
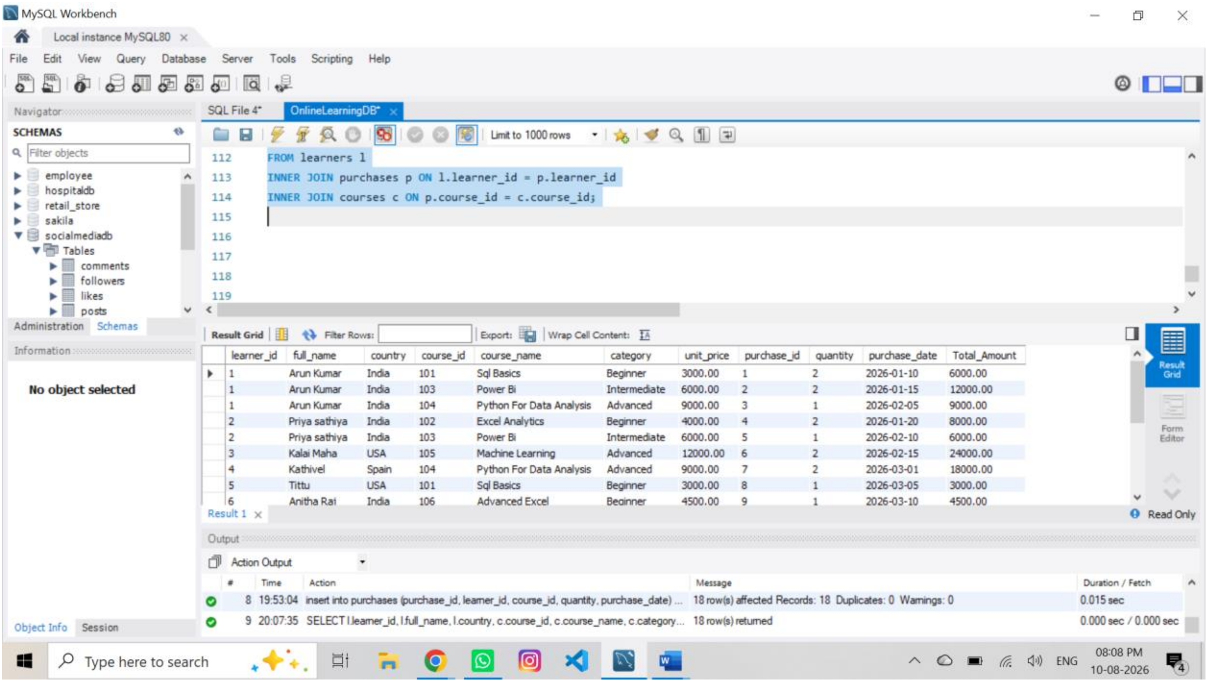


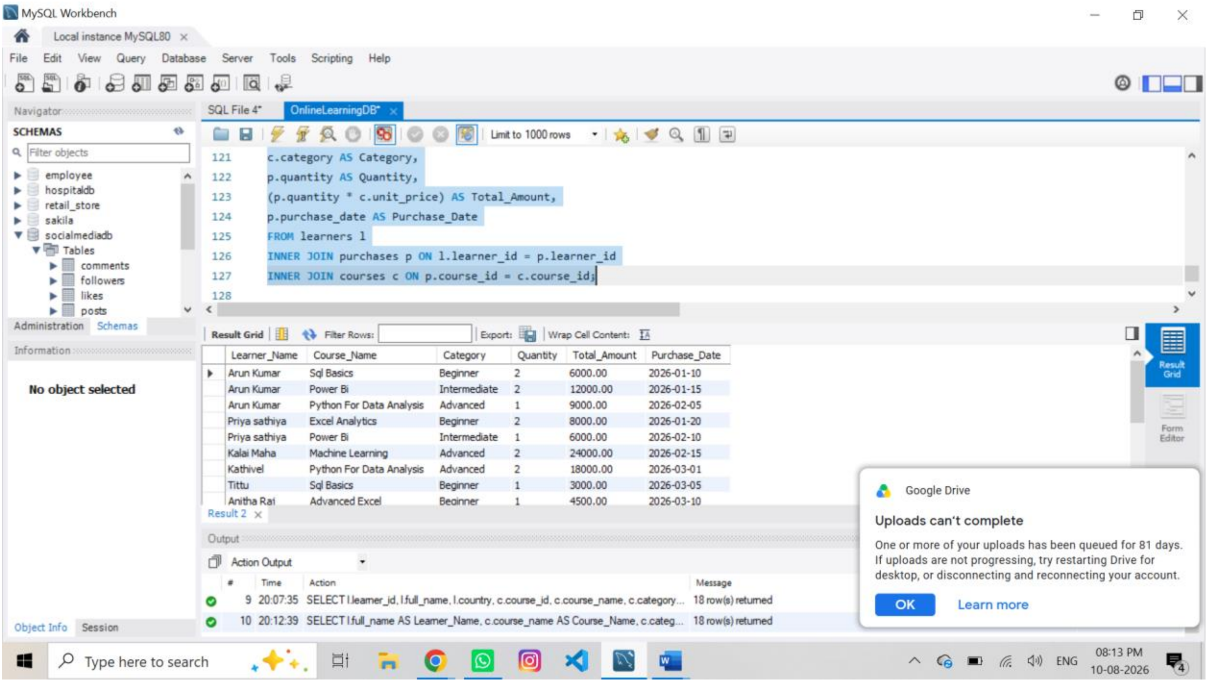
1. **Data Exploration Using Joins**

Use INNER, LEFT, and RIGHT JOIN to:

- **Combine learner, course, and purchase data**

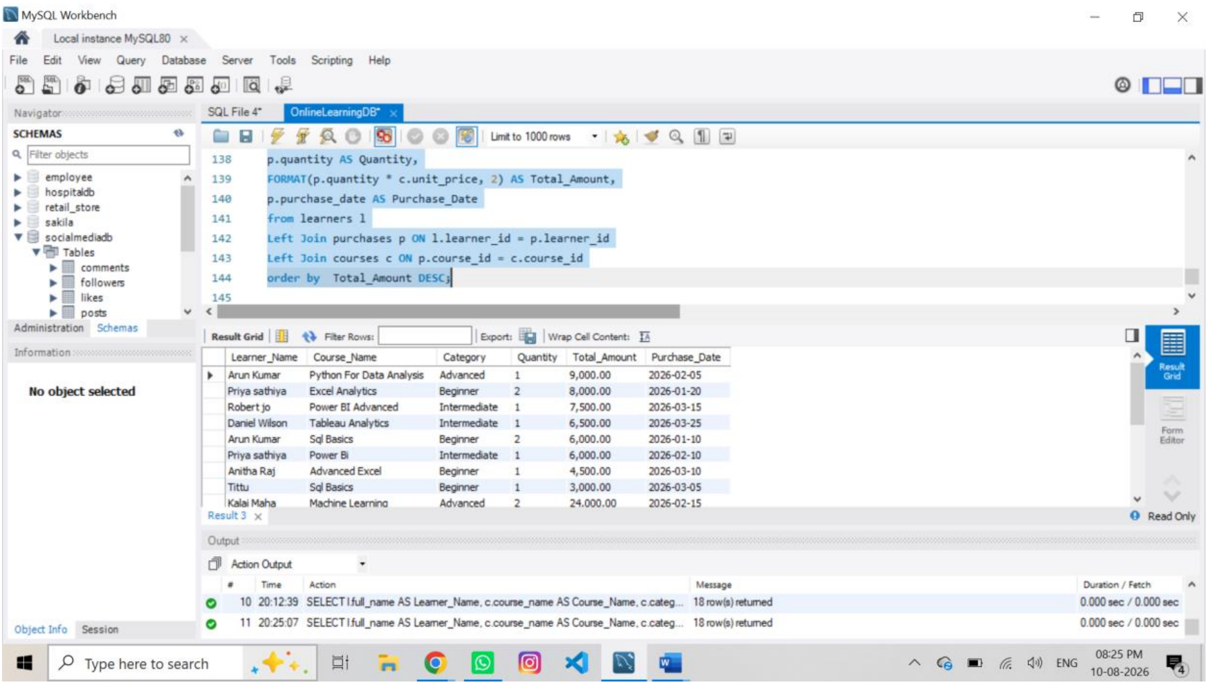


Display: Learner name, Course name, Category, Quantity, Total amount, Purchase date



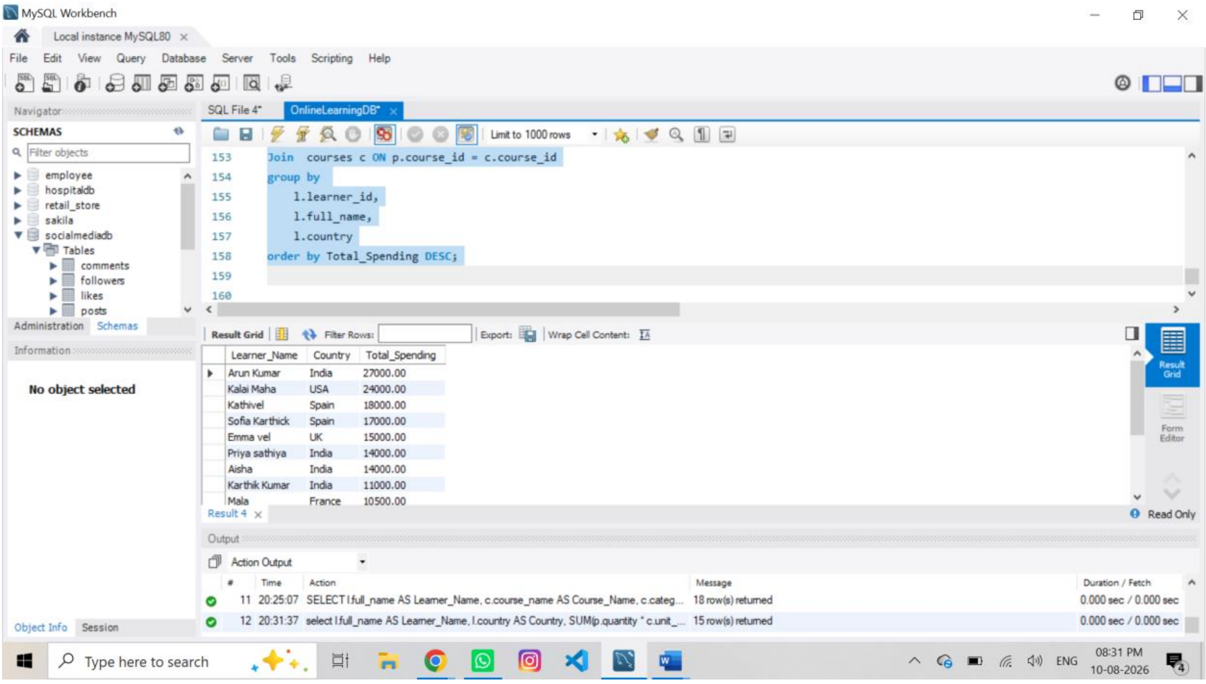
Presentation Requirements:

- **Format currency to 2 decimal places**
- **Use column aliases**
- **Sort by the highest total amount**

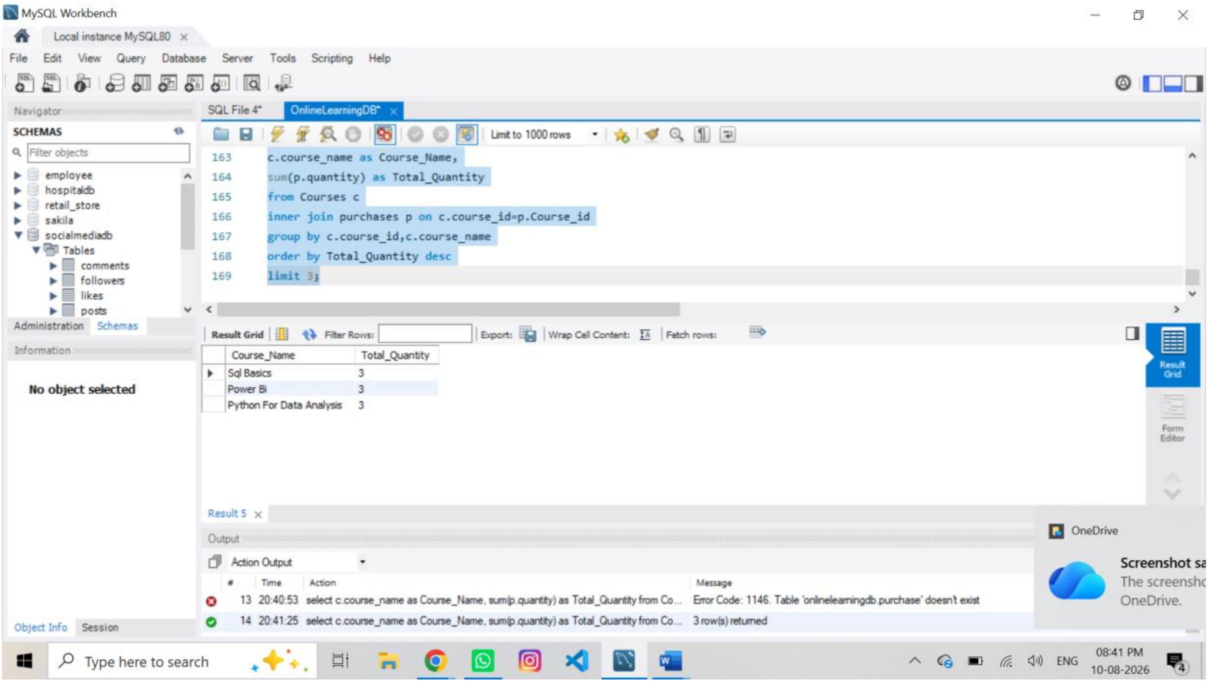


Core Analytical Queries (Q1–Q5)

Q1. Display each learner's total spending with their country.

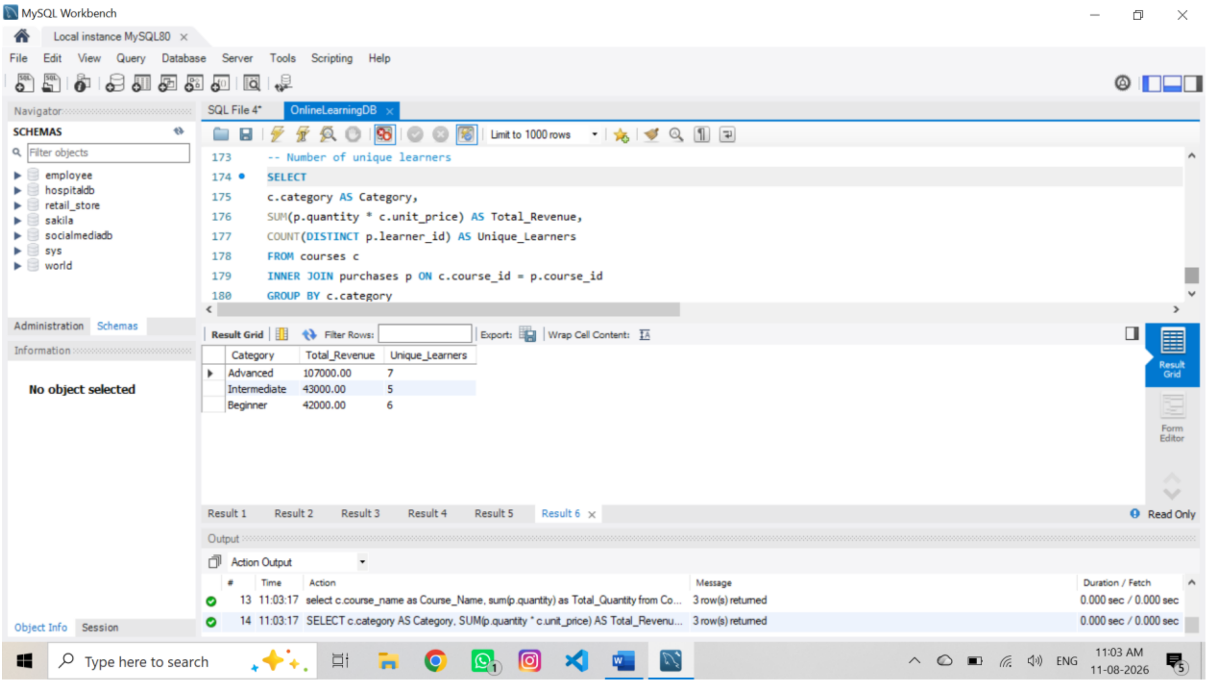


Q2. Find the top 3 most purchased courses by quantity

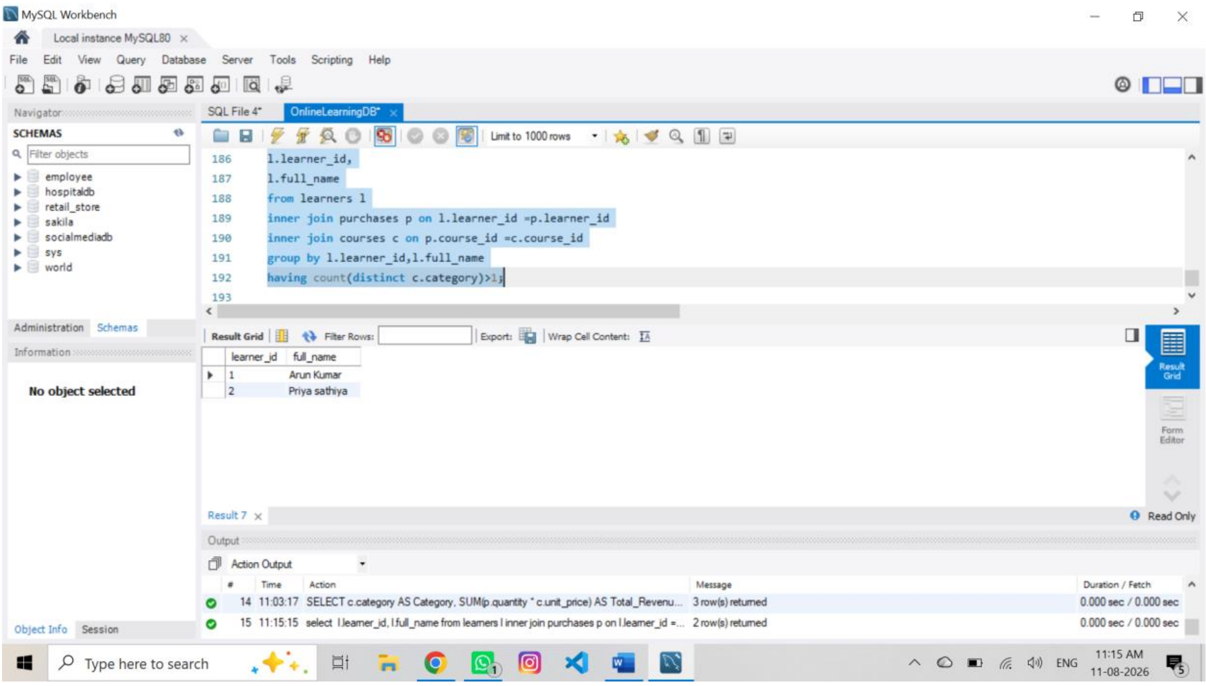


Q3. Show each category's:

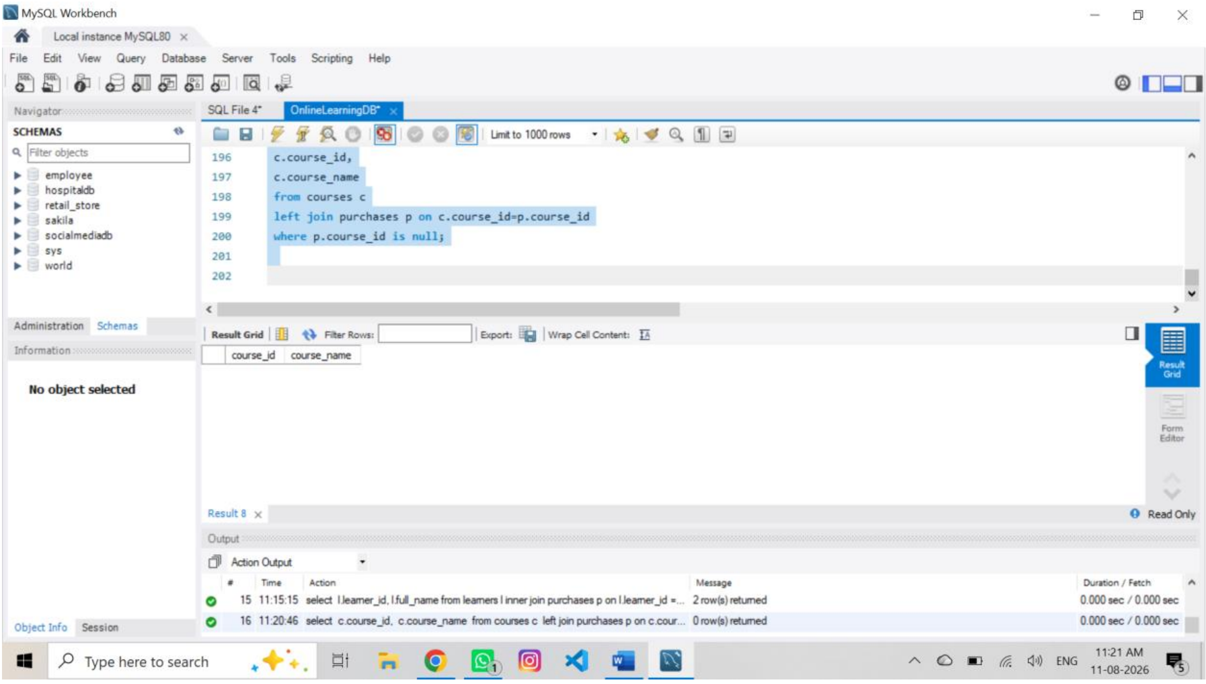
- Total revenue
- Number of unique learners



Q4. List learners who purchased from more than one category.

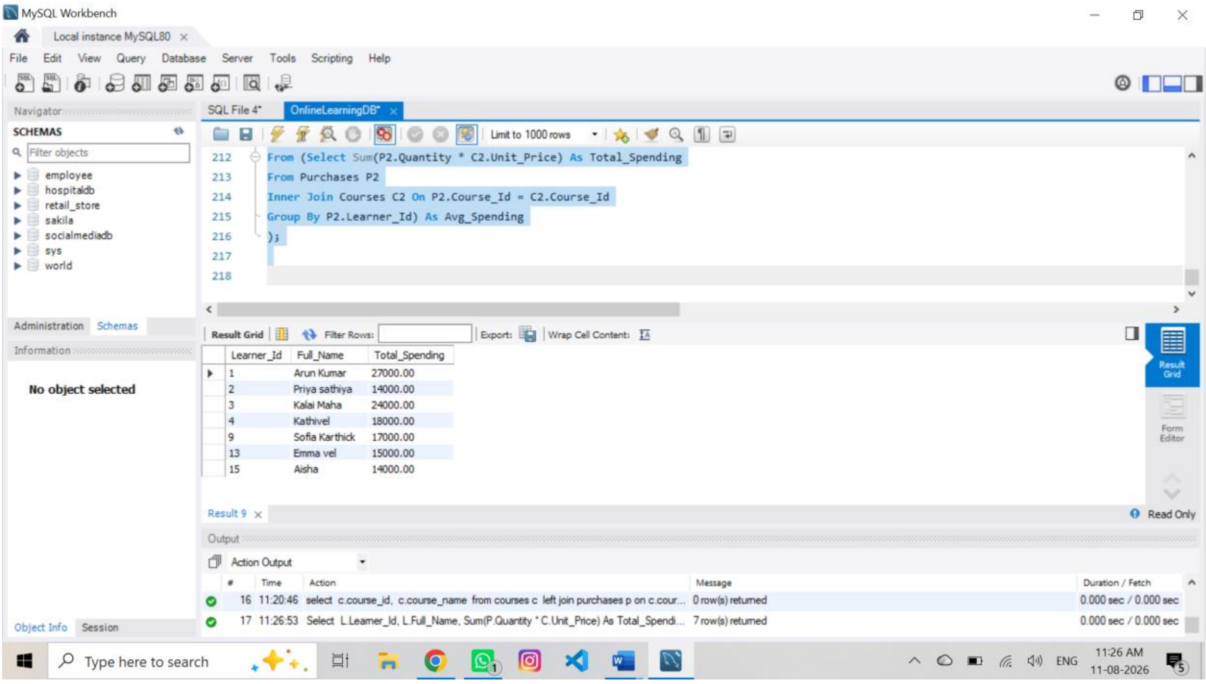


Q5. Identify courses never purchased.

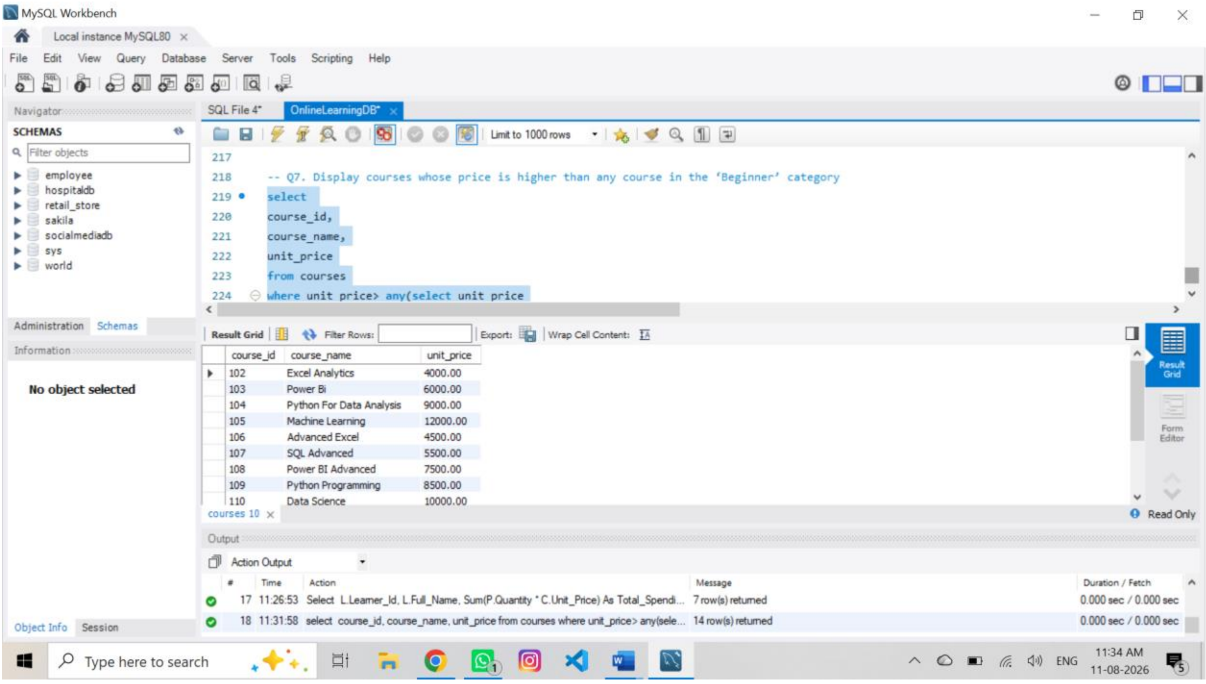


Subqueries & Correlated Subqueries

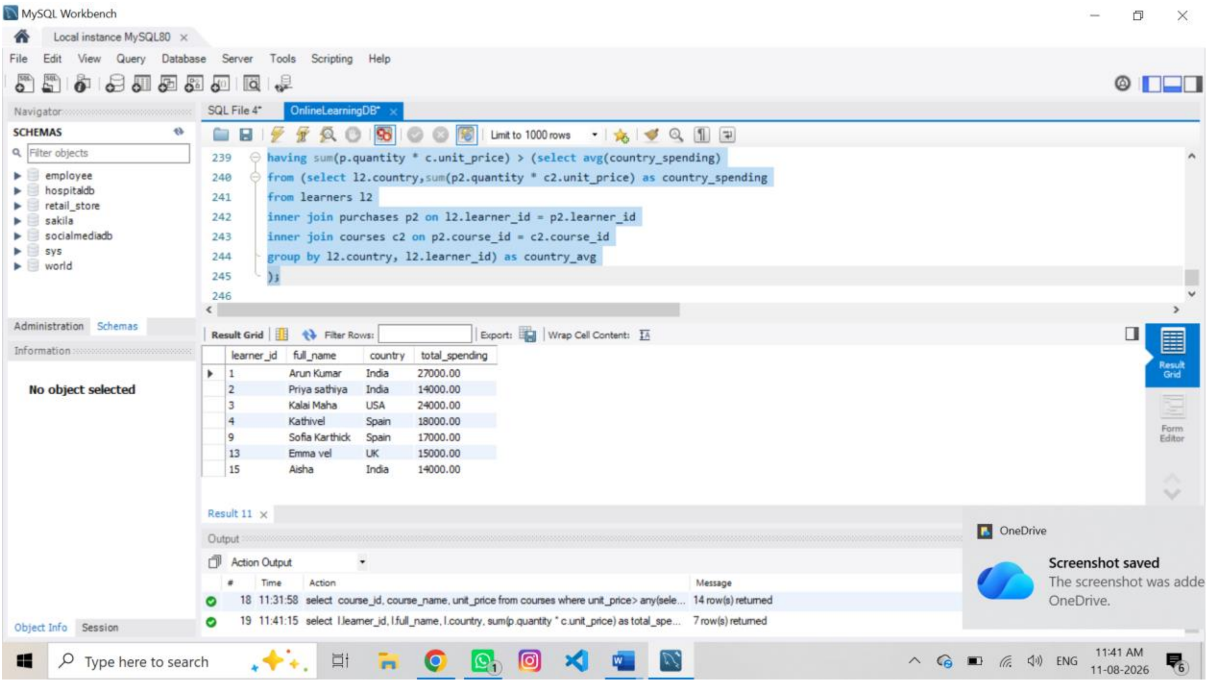
Q6. Find learners whose total spending is above the average learner spending.



Q7. Display courses whose price is higher than any course in the ‘Beginner’ category



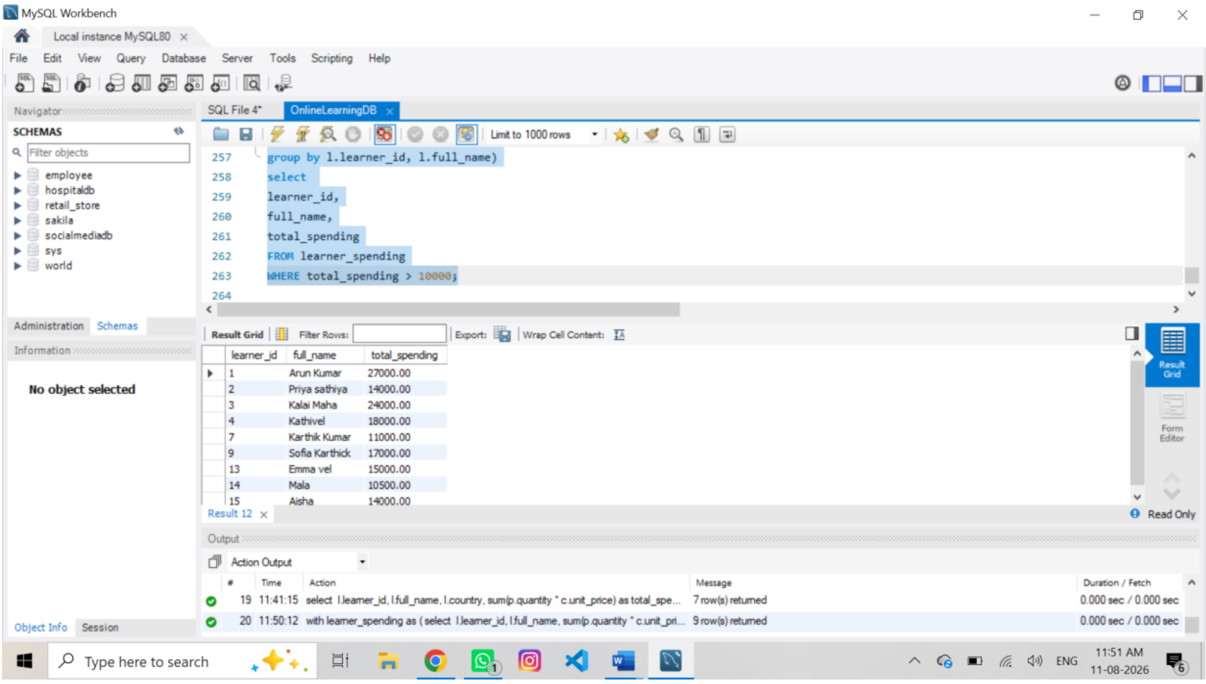
Q8 . Find learners who spent more than the average spending in their country.



5. CTE, CASE, View, and NULL Handling

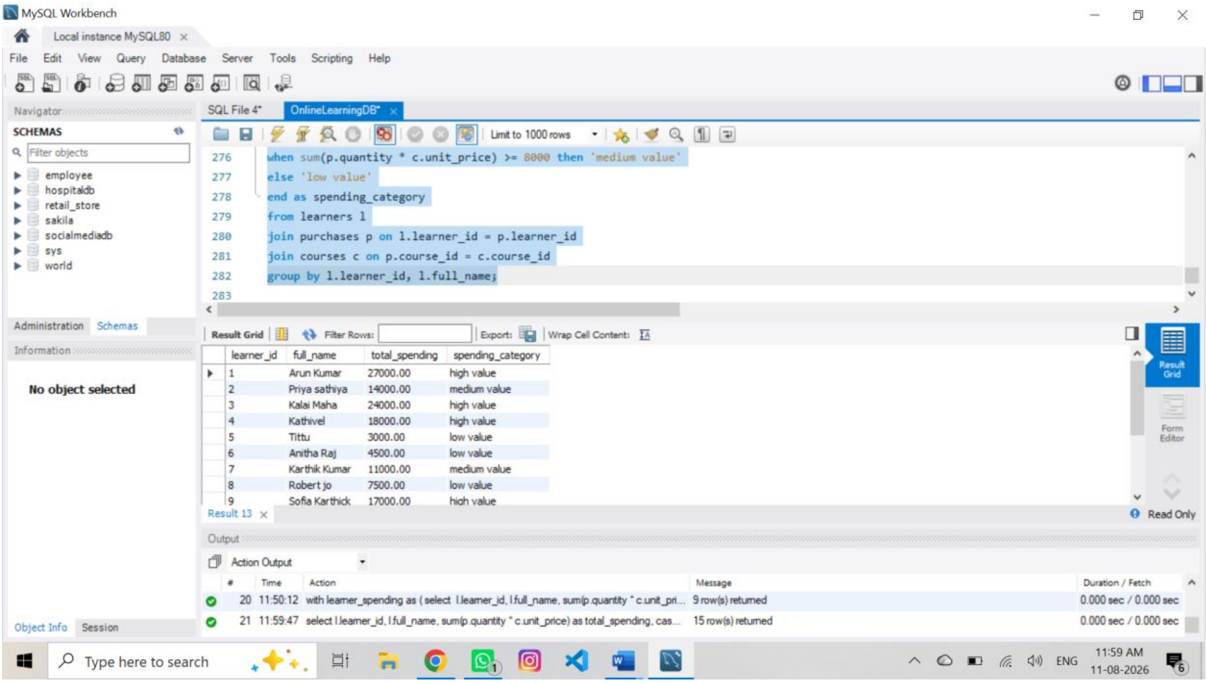
Q9. Use a CTE to calculate total spending per learner, then:

Display learners with spending above 10,000.



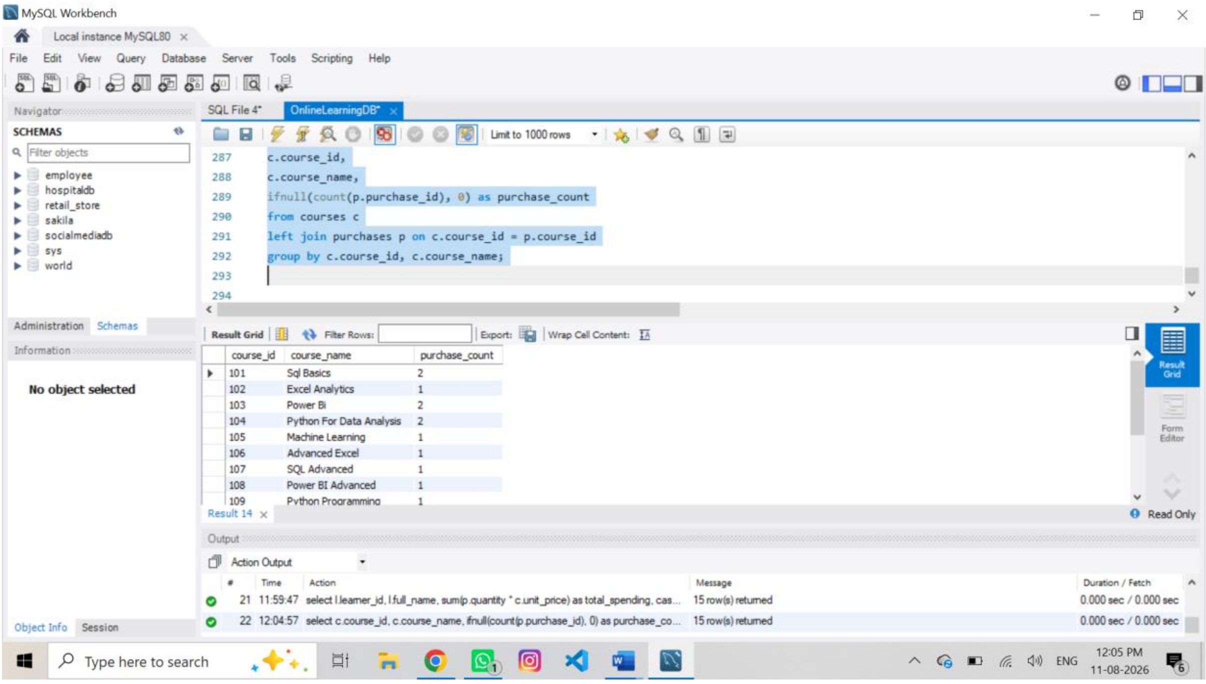
Q10. CASE Expression Classify learners based on spending:

- **Above 15,000 → "High Value",**
- **8,000–15,000 → "Medium Value",**
- **Below 8,000 → "Low Value".**



Q11 . NULL Handling

- **Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()**

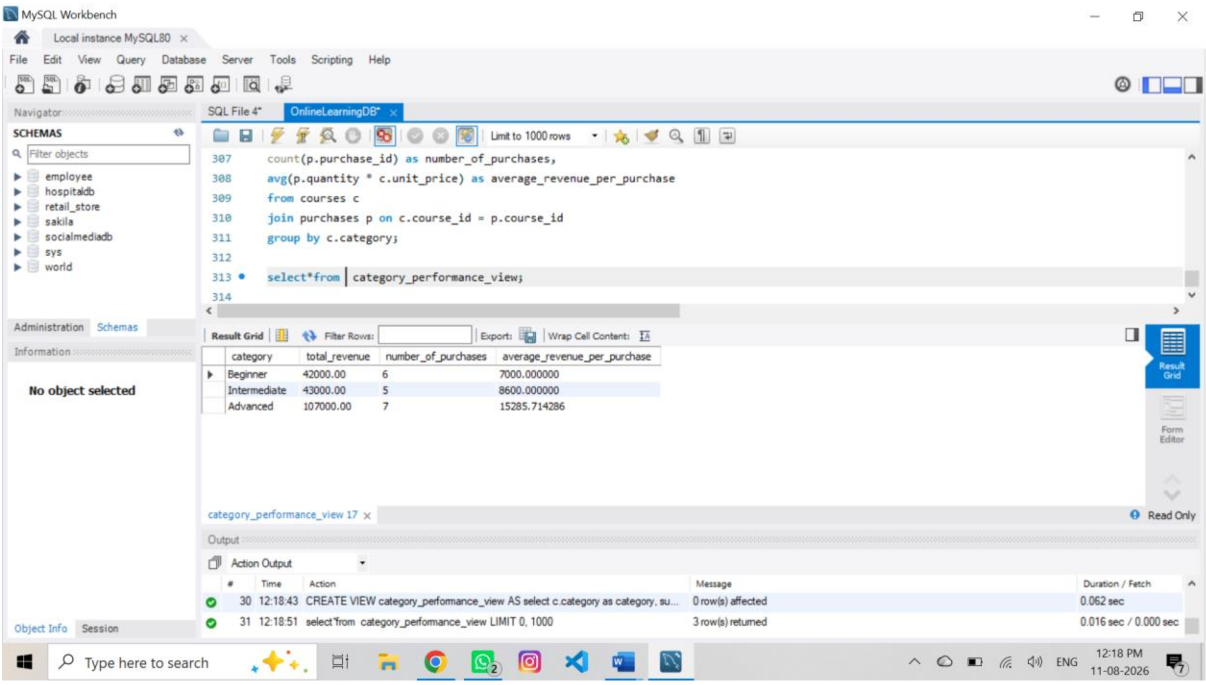


Q12 . View

- **Create a view:**

category_performance_view

- **Showing**
- **Category**
- **Total revenue**
- **Number of purchases**
- **Average revenue per purchase**



Conclusion :

The online learning platform database was successfully designed and managed using MySQL. SQL queries, joins, aggregate functions, CASE expressions, CTEs, NULL handling, and views were used to analyze learner purchases. The analysis provided useful insights into sales trends, learner spending behavior, course performance, and popular categories, helping the platform make better data-driven business decisions.